

1.2 解:

总灵敏度,

$$S = S_1 \times S_2 \times S_3$$

$$S = S_1 \cdot S_2 \cdot S_3 = 2 \text{ mm}/^\circ\text{C}$$

1.4 解:

设突变作用前后温差为 k , 又已知时间常数 $T=3s$

$$y(t) = k(1 - e^{-\frac{t}{T}})$$

$$y(t_1) = \frac{1}{3}k \quad \text{即} \quad \frac{1}{3} = 1 - e^{-\frac{t_1}{T}}$$

$$t_1 = -3 \ln \frac{2}{3} s = 1.22s$$

$$y(t_2) = \frac{1}{2}k \quad \text{即} \quad \frac{1}{2} = 1 - e^{-\frac{t_2}{T}}$$

$$t_2 = -3 \ln \frac{1}{2} s = 2.08s$$

则传感器输出温差 $\frac{1}{3}$ 的时间为 $1.22s$, 温差 $\frac{1}{2}$ 的时间为 $2.08s$

1.6 解:

压力差 $k = -20 \text{ MPa}$

$$y(t) = [25 - 20(1 - e^{-\frac{t}{0.5}})] \text{ MPa}$$

$$t_1 = 2T = 1s \quad y(t_1) = [25 - 20(1 - e^{-2})] \text{ MPa} = [25 - 20(1 - 0.135)] \text{ MPa} \\ = 7.7 \text{ MPa}$$

$$t_2 = 2s \quad y(t_2) = [25 - 20(1 - e^{-4})] \text{ MPa} = [25 - 20(1 - 0.018)] \text{ MPa} \\ = 5.36 \text{ MPa}$$

1.11 解:

$$\Delta L_1 = 0.2 \quad \Delta L_2 = 0 \quad \Delta L_3 = 0.02 \quad \Delta L_4 = 0.1 \quad \Delta L_5 = 0.1 \quad \Delta L_6 = 0.05$$

$$\Delta L_{\max} = \Delta L_1 = 0.2 \quad y_{FS} = 12.5 \text{ MPa}$$

$$\gamma_L = \pm \frac{\Delta L_{\max}}{y_{FS}} \times 100\% = \pm \frac{0.2}{12.5} \times 100\% = \pm 1.6\%$$

2) 由端点拟合得到的直线: $y = 1.97x + 0.23$

$$\Delta L_2 = 0.17 \quad \Delta L_3 = 0.16 \quad \Delta L_4 = 0.21 \quad \Delta L_5 = 0.02$$

$$\Delta L_{\max} = \Delta L_4 = 0.21$$

$$\gamma_L = \pm \frac{\Delta L_{\max}}{y_{FS}} \times 100\% = \pm 1.74\%$$

3) 最小二乘法求拟合直线

$$\sum x_i y_i = 1 \times 2.2 + 2 \times 4.0 + 3 \times 5.98 + 4 \times 7.9 + 5 \times 10.1 + 6 \times 12.05 = 182.54$$

$$\sum x_i = 21 \quad \sum x_i^2 = 91 \quad \sum y_i = 42.23$$

$$k = \frac{6 \sum x_i y_i - \sum x_i \sum y_i}{6 \sum x_i^2 - (\sum x_i)^2} = \frac{6 \times 182.54 - 21 \times 42.23}{6 \times 91 - 21^2} = \frac{208.41}{105} = 1.98$$

$$b = \frac{\sum x_i^2 \sum y_i - \sum x_i \cdot \sum x_i y_i}{6 \sum x_i^2 - (\sum x_i)^2} = \frac{91 \times 42.23 - 21 \times 182.54}{6 \times 91 - 21^2} = \frac{9.59}{105} = 0.09$$

$$y = 1.98x + 0.09$$

$$\Delta_i = y_i - (1.98x_i + 0.09)$$

$$\Delta_1 = 0.13 \quad \Delta_2 = -0.05 \quad \Delta_3 = -0.05 \quad \Delta_4 = -0.11 \quad \Delta_5 = 0.11 \quad \Delta_6 = 0.08$$

$$\Delta L_{\max} = 0.13 \quad \gamma_L = \pm \frac{\Delta L_{\max}}{y_{FS}} \times 100\% = \pm 1.08\%$$

1.12 解:

1) 迟滞误差: ① $y_{1FS} = 964.4 \quad \Delta H_{1\max} = 1.5$

$$\gamma_{H1} = \pm \frac{1}{2} \frac{\Delta H_{1\max}}{y_{1FS}} \times 100\% = \pm 0.08\%$$

② $y_{2FS} = 965.1 \quad \Delta H_{2\max} = 1.3$

$$\gamma_{H2} = \pm \frac{1}{2} \frac{\Delta H_{2\max}}{y_{2FS}} \times 100\% = \pm 0.07\%$$

③ $y_{3FS} = 965.7 \quad \Delta H_{3\max} = 1.8 \quad \gamma_{H3} = \pm \frac{1}{2} \frac{\Delta H_{3\max}}{y_{3FS}} \times 100\% = \pm 0.09\%$

2) 重复性误差

$$\Delta R_{\max 1} = 1.3 \quad \Delta R_{\max 2} = 1.3 \quad \Delta R_{\max} = 1.3$$

$$y_{FS} = 965.7 \quad \gamma_R = \pm \frac{\Delta R_{\max}}{y_{FS}} \times 100\% = \pm 0.13\%$$